

Polar Knowledge Index

NCPOR ke 40 saal ke polar expedition archive ko searchable, linked aur

trustworthy banane wala system — **aasaan Hinglish mein samjhaaya gaya**

“Every expedition. Every page. Every claim cited.”

 Archive

 Search

 AI with proof

 Human approval

IN English + हिन्दी

Ye document un logon ke liye hai jinhe technical background nahi chahiye.

Basics se shuru karke moderate level tak — har cheez diagram, flowchart, screenshot aur real data ke saath.

GitHub: github.com/saransh2809/SIH26063-Polar-Knowledge-Index

Is document mein kya hai?

1. Ek line mein — humne kya banaya
2. NCPOR kaun hai aur kya karta hai
3. Problem — data bikhra hua hai
4. Hamara idea — ek “smart librarian”
5. Poora system ek nazar mein (big picture)
6. Step 1 — Reports ikattha karna (Harvesting)
7. Step 2 — Purane scans padhna (Text + OCR)
8. Step 3 — Sab cheezon ko jodna (Linking)
9. Step 4 — Search: shabd aur matlab dono
10. Step 5 — AI se jawaab likhwana (RAG)
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12. Step 7 — Insaan ki approval + audit log
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14. Coverage view — kya missing hai
15. AI models — Gemini vs Local (free setup)
16. Accuracy kaise naapte hain
17. Technology — kaunsa tool kya karta hai
18. Abhi kya bacha hai
19. Judges ke possible sawaal + jawaab
20. Shabdkosh (Glossary)

BASICS

1 Ek line mein — humne kya banaya

Humne ek aisa system banaya hai jo **NCPOR ke already publish kiye hue purane expedition reports** ko khud ikattha karta hai, unhe padhta hai, har report ko uske **expedition, station, topic aur saal** se jodta hai — aur phir students, teachers aur outreach team ke sawaalon ka **jawaab sirf unhi reports se, page number ke saboot ke saath** deta hai.

 **Simple samjho:** Socho ek bahut bada library hai jisme 40 saal ki kitaabein hain, par na koi index hai, na koi librarian. Humne ek **smart librarian** banaya hai jo har kitaab padh chuka hai, har page yaad rakhta hai, aur jab aap kuch poochte ho to bolta hai: “ye raha jawaab — **Report 9, page 248** pe likha hai, khud dekh lo.” Aur agar library mein jawaab nahi hai, to **saaf bol deta hai “nahi pata”** — kuch bana ke nahi bolta.

Real numbers — jo system abhi chal raha hai

32

expedition reports
(DSpace se)

765

scientific papers
(har ek ka PDF)

837

NCPOR news posts
(2012 se aaj tak)

3,266

scanned pages
padhe gaye

6,377

searchable passages
(text ke tukde)

52

expeditions
coverage view mein

0

naye records jo NCPOR staff
ko type karne pade

31

automated tests
pass ho rahe hain

2 NCPOR kaun hai aur kya karta hai?

NCPOR = *National Centre for Polar and Ocean Research*, Goa. Ye Ministry of Earth Sciences (MoES) ke under ek research institute hai. Ye India ke **Antarctica, Arctic, Southern Ocean aur Himalaya** wale scientific expeditions chalata hai.



Har expedition ke baad kya banta hai?

Har saal scientists wahan jaate hain — glacier naapte hain, mausam record karte hain, chattano (rocks), samudra, jeev-jantu, aur insaan ke shareer par thand ka asar study karte hain. Wapas aakar ek **“Scientific Report”** chhapti hai jisme 20–40 research papers hote hain. Purani reports (1980s–2000s) ki **scanned PDFs** NCPOR ki ek website **DSpace** par padi hain.

Expedition

Ek research yatra. Jaise “9th Indian Expedition to Antarctica”. Hum ise code dete hain: **ISEA-9**.

Station

Antarctica/Arctic mein India ka research ghar. Jaise **Maitri, Bharati, Himadri**.

Report → Paper → Page

Ek report mein kayi papers; har paper ke kayi pages. Hum har page tak track karte hain.

Simple samjho: NCPOR ke paas khazana hai — 40 saal ka Indian polar science. Problem ye nahi ki data nahi hai. Problem ye hai ki ye khazana **bandh almari** mein hai: scanned PDFs jinhe na search kar sakte ho, na aapas mein jod sakte ho.

Problem — data 6 jagah bikhra hua hai

NCPOR ka material kam se kam **6 alag-alag systems** mein hai, aur inke beech koi connection nahi hai:



Koi link nahi: dataset ko report ka pata nahi,
report ko photos ka pata nahi, news ko dono ka pata nahi

Ek student ki pareshani — real example

Sawaal: "India ne Bharati station par kya research kiya?"

Aaj ka raasta: main website (station page) → DSpace (reports, scanned, search nahi hota) → NPDC (datasets) → Google Scholar (papers) → YouTube (videos)... **5 jagah, aur kahin bhi koi link nahi.** Zyada tar log thak ke internet ki unverified jaankari use kar lete hain.

Kis-kis ko dikkat hoti hai?

Kaun	Dikkat	Nateeja
NCPOR outreach team	Har news post zero se likhni padti hai, background facts haath se dhoondhne padte hain	Outreach irregular
Teachers	India ke apne polar research par koi teaching material nahi	Bachche videshi examples padhte hain
Students	Simple sawaal ka bharosemand jawaab nahi milta	Galat internet content
Researchers	Purane expedition ka kaam scanned PDFs mein band hai	Kaam dobara hota hai
Journalists	Jaldi verified fact chahiye	News mein galtiyan

💡 **Simple samjho:** Asli problem "website nahi hai" nahi hai — asli problem hai **"links nahi hain"**. Humara system wahi missing links banata hai.

Hamara idea — ek “smart librarian”

Hum NCPOR se koi naya data type karwane ko nahi bolte. Jo already publish ho chuka hai, wahi utha ke usko **padhte, jodte, dhoondhne layak banaate** hain — aur AI se likhwaaya gaya har shabd **page ke saboot** ke saath aata hai.

Hamare 4 vaade (promises)

1. Zero naya data entry

System khud DSpace aur NCPOR news feed se records utha leta hai. Staff ko kuch type nahi karna.

2. Sab kuch juda hua

Har report, paper aur news → kis expedition, kis station, kis topic, kis saal ka hai.

3. Har line ka saboot

AI ka har sentence batata hai ki wo kis report ke kis page se aaya. Saboot nahi → jawaab nahi.

4. Insaan ki final approval

AI sirf draft banata hai. Ek naam wala NCPOR reviewer approve kare tabhi public hota hai.

Doosri SIH teams se hum alag kyon hain?

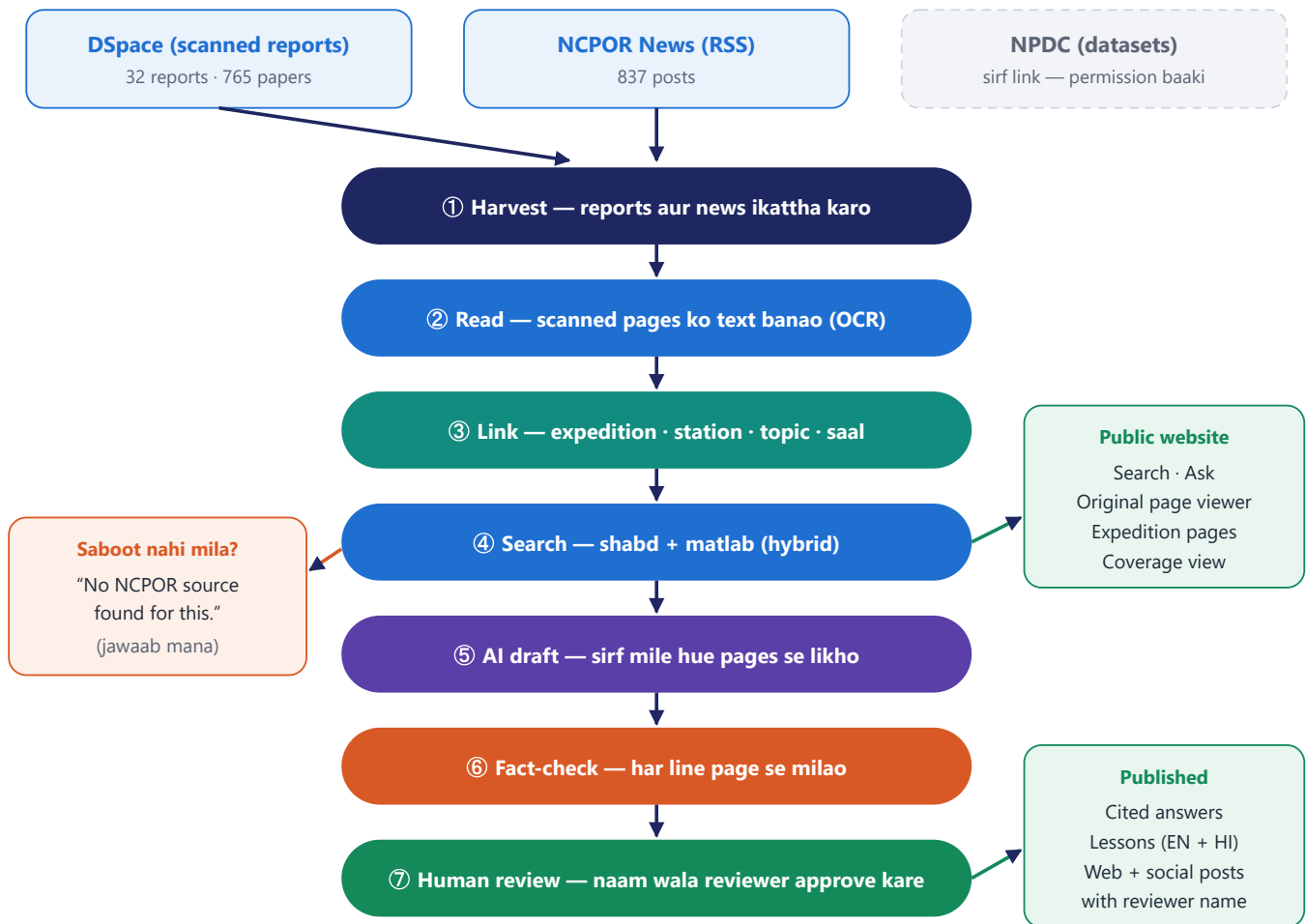
Is problem statement par kai teams ne “AI portal” banaye hain. Hum 4 cheezon mein alag hain:

Cheez	Aam SIH prototype	Polar Knowledge Index
NCPOR ka asli archive use karta hai	✗ seed/dummy data	✓ 32 real reports
Staff ko naya data type karna padta hai	✗ upload forms	✓ automatic harvest
Har sentence ka fact-check, galat line approval rokta hai	✗	✓
Accuracy naapi gayi (measured)	✗ sirf claim	✓ verified test set
Coverage view — kya missing hai	✗	✓

💡 **Simple samjho:** Baaki teams kehti hain “humara AI accurate hai”. Hum kehte hain “humne **naapa** hai, aur ye raha number” — aur har jawaab ke saath **original scanned page** dikhate hain. Judges ke liye ye sabse bharosemand cheez hai.

5 Poora system ek nazar mein

Ise ek factory ki assembly line ki tarah socho. Kachcha maal (NCPOR ke reports) ek taraf se aata hai, aur doosri taraf se **search, jawaab, lessons aur announcements** nikalte hain — beech mein 7 steps.

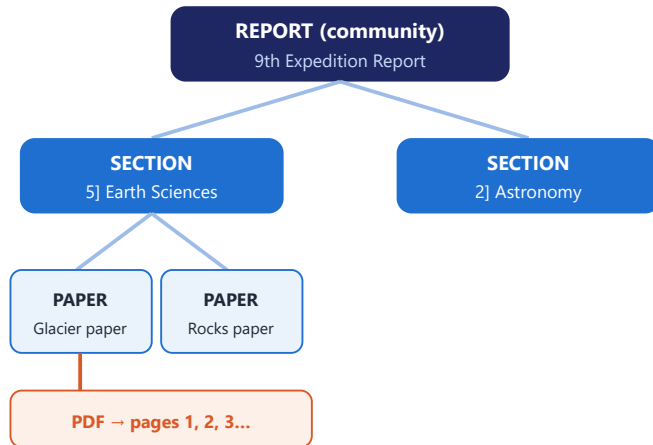


Steps ①–④ “archive” hissa hain (bina AI ke bhi chalte hain). Steps ⑤–⑦ “AI + insaan” hissa hain.

6 Reports ikattha karna (Harvesting)

Harvesting ka matlab hai doosri website se records apne aap copy karna — bilkul waise jaise khet se fasal kaatte hain. Humara program DSpace aur NCPOR news feed se ye kaam karta hai.

DSpace ka structure (ek tree jaisa)



Hum ek “achhe mehmaan” ki tarah aate hain

- **robots.txt padhte hain** — har website ek file rakhti hai jo batati hai ki bots kya le sakte hain. Hum uska paalan karte hain.
- **1 second mein sirf 1 request** — NCPOR ka server overload na ho.
- **Har file ek hi baar download** — copy apne computer par cache mein, dobara nahi maangte.
- **Original wahin rehta hai** — hum PDF dobara publish nahi karte, sirf link + padha hua text rakhte hain.
- **NPDC ko nahi chhuua** — wahan ke datasets ke liye NCPOR ki permission chahiye.

OAI-PMH kya hai? Libraries ka ek standard “catalogue copy” protocol. DSpace mein ye pehle se ON tha — isliye humne bina scraping ke saaf-suthre records le liye.

Harvest ka result

32

report communities

765

papers (metadata)

264

PDFs download (10 reports)

837

news posts

💡 **Simple samjho:** Dobara chalao to system kuch bhi dobara download nahi karta aur ek bhi duplicate record nahi banata — humne test kiya: **0 network requests, 0 naye duplicates.**

7 Purane scans padhna (Text layer + OCR)

Scanned PDF asal mein **photo** hoti hai — computer ke liye usme “shabd” nahi, sirf pixels hote hain. Use padhne ke 2 tareeke:

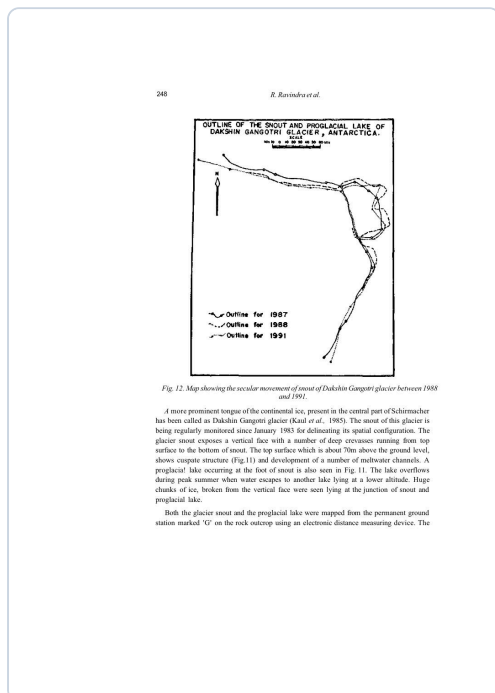
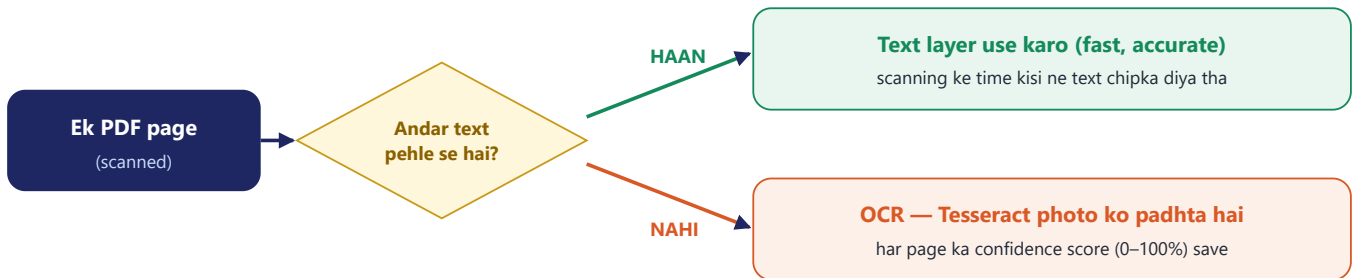


Fig. 12. Map showing the secular movement of snout of Dakshin Gangotri glacier between 1988 and 1991.

A more prominent tongue of the continental ice, present in the central part of Schirmacher has been called as Dakshin Gangotri glacier (Kail et al., 1985). The snout of this glacier is being regularly monitored since January 1985 for delineating its spatial configuration. The glacier snout exposes a vertical face with a number of deep crevasses running from top surface to the bottom of snout. The top surface which is about 7m above the ground level, shows cusped structure (Fig. 11) and development of a number of meltwater channels. A proglacial lake occurring at the foot of snout is also seen in Fig. 11. The lake overflows during peak summer when water escapes to another lake lying at a lower altitude. Huge chunks of ice, broken from the vertical face were seen lying at the junction of snout and proglacial lake.

Both the glacier snout and the proglacial lake were mapped from the permanent ground station marked 'G' on the rock outcrop using an electronic distance measuring device. The

Asli scan: 9th Expedition report, printed page 248 — Dakshin Gangotri glacier snout ka map

Har page ke saath kya save hota hai?

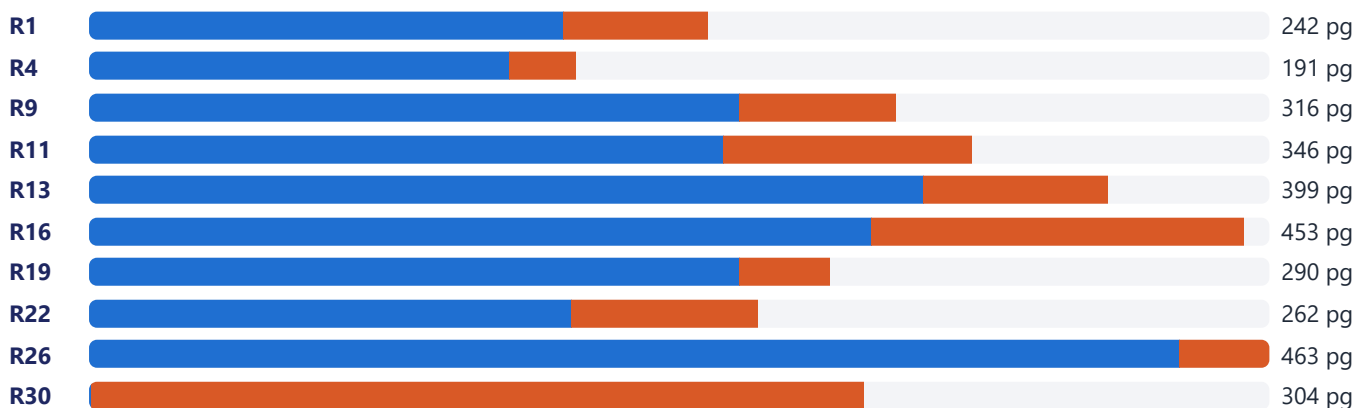
- **Text** — searchable
- **Kahan se aaya** — text layer ya OCR
- **OCR confidence** — kitna bharosa
- **Printed page number** — jaise “248” (PDF ka page 10 ≠ report ka page 248!)
- **Page ki image** — taaki “View original page” dikha sakein

Simple samjho: Har jawaab ke saath hum ye scan dikhate hain. Agar OCR ne galti ki ho, to padhne wala khud asli page dekh ke check kar sakta hai.

Chunks: har page ke text ko chhote tukdon ($\approx 1,500$ characters) mein kaata jaata hai. Ek tukda kabhi do pages mein nahi phailta — taaki har citation **ek exact page** bataye. Total: **6,377 chunks**.

Real data — 10 reports, 3,266 pages

■ Text layer se ■ OCR se



Finding: 10 mein se 9 reports ke 68–92% pages mein pehle se text layer tha. Report 30 poora OCR karna pada — average confidence **90%**. Saare 10 reports ~2 minute mein padh liye gaye (laptop par).

Sab cheezon ko jodna (Linking)

Ye humara **sabse important step** hai — yahi “missing links” banata hai. Har item (report, paper, news) ko 4 cheezon se joda jaata hai:



Link kaise banta hai? Pehle rules, phir AI

Tareeka	Example	Status
Report ka title padho	"Scientific Report Of Ninth Indian Expedition" → ISEA-9 ("Twenty First" → 21 bhi samajhta hai)	Confirmed
NCPOR ka section	Paper "5] Earth Sciences" section mein hai → topic Geology	Confirmed
Printed header	Page 1 par "Scientific Report, 1994" → year 1994	Confirmed
Keywords	Text mein "Maitri" 2+ baar → station Maitri	Unconfirmed
AI (fallback)	Jahan rules fail hon, AI fixed list mein se topic chunta hai + saboot quote	Unconfirmed

Rule simple hai: jo NCPOR ne khud likha (title, section, printed page) → **confirmed**. Jo humne ya AI ne andaaza lagaya → **unconfirmed**, jab tak ek curator (NCPOR staff) "Confirm" ya "Reject" na dabaye. Har link ke saath **wajah (evidence)** bhi save hoti hai.

32/32

reports ka expedition link

765/765

papers ka expedition link

3,616

links rules se

65

links AI se (unconfirmed)

Ek galti jo humne pakdi: pehle version ne news ki line "The **first** team of the 45th ISEA" ko "Expedition 1" samajh liya, aur "42nd **Chinese** Antarctic Expedition" ko India ka ISEA-2! Humne rule sakht kiya — number ke turant baad "(Indian) (Scientific) Expedition" hona chahiye — aur in galtiyan ke **tests** likhe taaki dobara na ho.

Bonus: news se humein **expeditions 33–46** ka pata chala jinki report DSpace mein hai hi nahi — ye coverage view mein dikhta hai (chapter 14).

9 Search — shabd aur matlab, dono

Purani reports alag shabd use karti hain. Student likhta hai **“glacier kitna pighla”**, report mein likha hai **“ablation”** ya **“secular movement”**. Normal search ye match nahi kar paata. Isliye hum **2 search** ek saath chalaate hain:

Keyword search (shabd)

Exact shabd dhoondhta hai — jaise “Maitri”, “ISEA-9”, instrument codes.

Tool: PostgreSQL full-text search

✓ Exact naam kabhi miss nahi · ✗ Synonyms nahi samajhta

Meaning search (matlab)

Har text ko **768 numbers ki list (embedding)** mein badalta hai. Milte-julte matlab = milte-julte numbers.

Tool: multilingual-e5 model, laptop par chalta hai

✓ “moving” ≈ “movement” · ✗ Exact codes kabhi miss

Simple samjho: Embedding ko aise samjho: har paragraph ko ek **naksha (map) par point** bana diya. “Glacier pighalna” aur “glacier ablation” map par paas-paas aate hain, “cricket match” door. Sawaal ka point banao, aur sabse paas wale paragraphs utha lo.

Dono ko milana — “Hybrid” (RRF)

Dono searches apni-apni ranking deti hain. **Reciprocal Rank Fusion** ek simple formula hai: jo paragraph kisi bhi list mein upar hai, wo final list mein bhi upar aata hai. Koi tuning nahi chahiye.

Jawaab kab MANA karna hai? (Refusal)



Ye numbers real hain — 7 sawaalon par naapa. Archive wale sawaal 0.84–0.88, bahar ke sawaal 0.77–0.80. Isliye line 0.82 par khinchi.

Search the archive

☒ Hybrid (recommended)
 ☐ Exact words
 ☐ Meaning

15 passages for “Dakshin Gangotri glacier movement”. These passages are the reports' own words, not AI text.

Monitoring of Icebergs in Antarctic Waters and a Note on the Secular Movement of Dakshin Gangotri Glacier
09] Scientific Report Of Ninth Indian Expedition To Antarctica

248
R. Ravindra et al.
Fig. 12. Map showing the secular movement of snout of Dakshin Gangotri glacier between 1988 and 1991.
A more prominent tongue of the continental ice, present in the central part of Schirmacher has been called as Dakshin Gangotri glacier (Kaul et al., 1985). The snout of this glacier is being regularly monitored since January 1983 for delineating its spatial configuration. The glacier snout exposes a vertical face with a number of deep crevasses running from top surface to the bottom of snout. The top surface which is about 70m above the ground level, shows cusped structure (Fig.11) and development of a number of meltwater channels. A proglacial lake occurring at t...

Page 10 (printed p. 248) [Text from the PDF](#) [View original page](#)

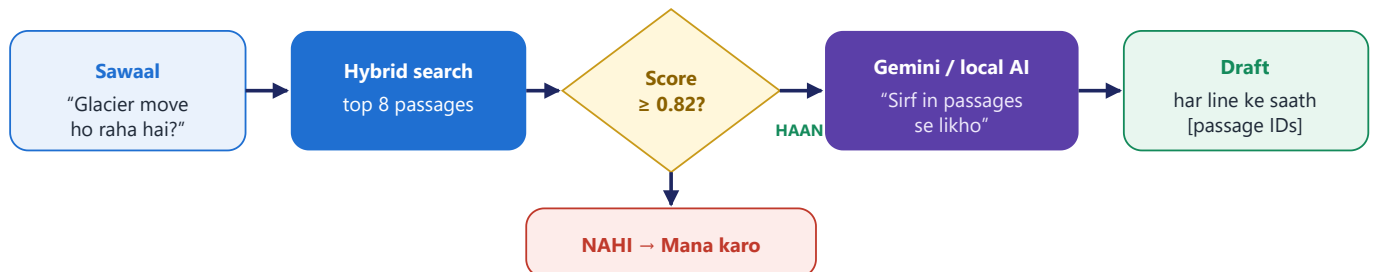
Monitoring of Icebergs in Antarctic Waters and a Note on the Secular Movement of Dakshin Gangotri Glacier
09] Scientific Report Of Ninth Indian Expedition To Antarctica

References

Asli search: “Dakshin Gangotri glacier movement” → pehla result 9th Expedition ka glacier paper, printed page 248. “These passages are the reports' own words, not AI text.”

AI se jawaab likhwaana (RAG)

RAG = *Retrieval-Augmented Generation*. Bhaari naam, simple idea: **pehle dhoondo, phir sirf usi se likho**. AI ko apni yaaddasht se kuch bolne ki ijaazat nahi — sirf jo pages humne diye, unhi se.



AI ko diye gaye sakht niyam (rules)

- Har fact wali line ke saath un passages ke **IDs** do jo use support karte hain (1–3 hi, saare nahi).
- Jo passages mein nahi likha — numbers, dates, naam — **mat jodo**, chahe tumhe pata ho.
- OCR wale text mein agar number saaf nahi dikhta, to **number mat likho**.
- Kisi pehchaane jaa sakne wale insaan ki **health / psychology** par mat likho.
- Agar passages jawaab nahi dete — "answerable: false" bolo, **kuch mat banao**.

Safety net: agar AI koi aisa passage ID likh de jo humne diya hi nahi tha, system wo citation **khud hata deta hai**. Aur agar AI kahe "ye line fact nahi hai, check mat karo" — system nahi maanta (ye humne real run mein pakda aur fix kiya).

3 tarah ke drafts

Cited answer

Student ke sawaal ka 2–5 line ka jawaab, har line page se judi.

Lesson unit

Class 6–12 ke liye: aasaan + advanced explainer, starter sawaal, activity, 5 quiz sawaal, teacher notes.

Announcement

Expedition par website post + X, Facebook, Instagram posts — sirf confirmed records se.

Hindi: approved English draft ka Hindi anuvaad banta hai, uski bhi alag se jaanch aur approval hoti hai.

Fact-checker — har line ki jaanch

Draft banne ke baad ek **doosra AI pass** har sentence ko uske cited passage se milata hai — jaise teacher copy check karta hai — aur 3 mein se ek label lagata hai:

✓ Supported

Har detail (naam, number, date, jagah) passage mein likhi hai.

! Partly supported

Main baat sahi, par koi detail passage mein nahi ya zyada badha-chadha ke.

✗ Not supported

Passage mein nahi likha, ya ulta likha hai. **Approval block!**

NCPOR Polar Knowledge Index Search Ask a question Expeditions Coverage Lessons & stories Accuracy हिन्दी Staff

Overview **Review queue** Draft new content Confirm links Audit log Signed in as **Demo Curator** (curator) [Sign out](#)

Lesson · English · drafted by Demo Curator · version 1

Observing Glacier Movement: The Dakshin Gangotri Glacier in Antarctica

Auto-checked ✓ 21 supported 2 partly supported ✗ 4 not supported

Explainer (introductory)

✗ Not supported Edit Delete

Glaciers are large bodies of ice that move slowly.
Checker: No valid citation to an archive passage.

✓ Supported Show sources (2) Edit Delete

Indian scientists monitor the Dakshin Gangotri (DG) Glacier in Antarctica to understand how continental ice changes over time (3221, 6118).
Checker: The passages state that Indian Antarctic Expedition monitors the Dakshin Gangotri glacier to understand the dynamics and temporal fluctuations of continental ice.

✓ Supported Show sources (1) Edit Delete

This glacier is a part of the larger polar ice sheet (6118).
Checker: Passage 6118 states that the DG Glacier snout is a 'descending part of the Polar ice sheet'.

! Partly supported Hide sources (2) Edit Delete

Scientists observe whether the glacier's front, called the snout, advances (moves forward) or recedes (moves backward) (3221, 6135).
Checker: The passages mention observing 'advance' and 'recession' of the snout, but do not explicitly define these terms as 'moves forward' or 'moves backward'.

[3221] Monitoring of Icebergs in Antarctic Waters and a Note on the Secular Movement of Dakshin Gangotri Glacier, page 11 (printed p. 249) [open page](#)
Monitoring of Icebergs ...
249

Decision
4 sentence(s) are not supported by their sources. Edit or delete them before approving.
Only a reviewer can approve or publish.
[Re-run checker](#)

Review history
No decisions yet.

Asli review screen: Class 11 glacier lesson — 21 supported, 2 partly, 4 not supported. Pehli line "Glaciers are large bodies of ice that move slowly." RED hai kyunki ye general knowledge hai, NCPOR ke passages mein nahi.

Real example — red flag:

"Glaciers are large bodies of ice that move slowly." → ✗ **Not supported**. Baat sach hai, par diye gaye NCPOR pages mein nahi likhi. Humara system sirf **saboot wali** baat chhapti hai. Reviewer ise delete karega ya source jodega.

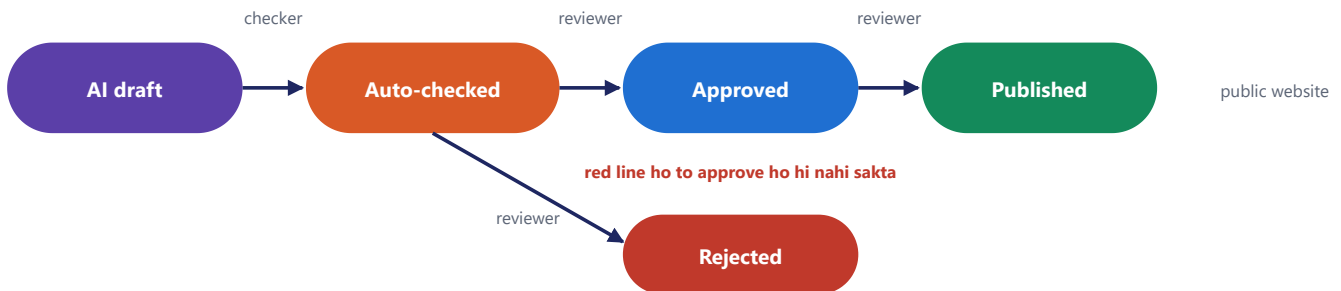
Real example — Hindi galti:

"austral summer" (dakshini golaardh ki garmi) ka anuvaad AI ne "ऑस्ट्रेलियाई गर्मी" (Australia ki garmi) kar diya — aur checker ne pass kar diya! Humne glossary jodi, ab sahi aata hai: "दक्षिणी गोलार्ध की गर्मी".
Seekh: AI checker bhi galti karta hai — isliye insaan ki approval zaroori hai.

💡 **Simple samjho:** Checker ek **chhanni (filter)** hai, judge nahi. Ye reviewer ka kaam aasaan karta hai — red lines pe dhyaan do — par final faisla insaan ka hai.

12 Human approval + audit log

AI ka likha kuch bhi **tab tak public nahi hota** jab tak ek naam wala reviewer approve aur publish na kare. Har draft ek safar tay karta hai:



Kaun kya kar sakta hai? (Roles)

Role	Kya karta hai
Public	Sirf padhta hai — login nahi, likh nahi sakta
Curator	Machine links confirm/reject karta hai, drafts banwata hai
Reviewer	Approve, reject, publish, galti ho to correction
Admin	Sab kuch

Audit log — “CCTV camera”

Har action record hota hai: kisne, kab, kya kiya — login, draft bana, line edit hui, approve hua.

Append-only: database khud ek नियम lagata hai ki purani entry ko **na badla jaa sakta, na mitaaya** — chahe code mein bug ho. Test se verify kiya gaya.

Galti publish ho gayi? Chupke se badalte nahi — naya version banta hai “correction note” ke saath, purana history mein rehta hai.

answer · English · drafted by Demo Curator · version 1

Has India measured how the Dakshin Gangotri glacier is moving?

Auto-checked ✓ 4 supported ! 0 partly supported ✗ 0 not supported

Answer

✓ Supported [Show sources \(3\)](#) [Edit](#) [Delete](#)

India has regularly monitored the Dakshin Gangotri Glacier since the austral summer of 1983 to track its secular and temporal movement.

Checker: All details about monitoring the Dakshin Gangotri Glacier since 1983 for its secular and temporal movement are explicitly stated in the passages.

✓ Supported [Show sources \(3\)](#) [Edit](#) [Delete](#)

The snout boundary of the glacier and its proglacial lake have been mapped using electronic distance measurement units.

Checker: The passages explicitly state that the snout boundary and proglacial lake were mapped using electronic distance measurement units.

Asli answer draft: “Has India measured how the Dakshin Gangotri glacier is moving?” — 4 mein se 4 lines ✓ supported, har ek ka source dekh sakte ho. Abhi “Auto-checked” — reviewer ka intezaar.

13 Website ka tour

PRODUCT

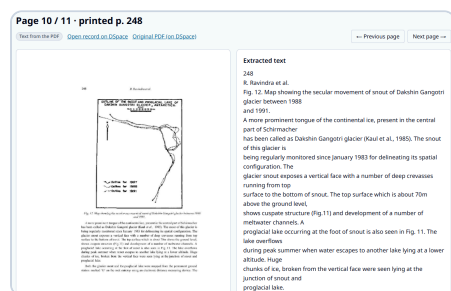
Public hissa (koi bhi dekh sakta hai)

Page	Kaam
Search	Poore archive mein dhoondo (hybrid / exact / meaning)
Ask a question	Sawaal pucho → matching passages + approved jawaab, ya “mana”
Page viewer	Scan aur uska text side-by-side
Expeditions	Har expedition ka ek page: report, papers, news, stories
Coverage	Kis expedition ki report hai, kiski nahi
Lessons & stories	Approved content, English/हिन्दी
Accuracy	Evaluation ke numbers

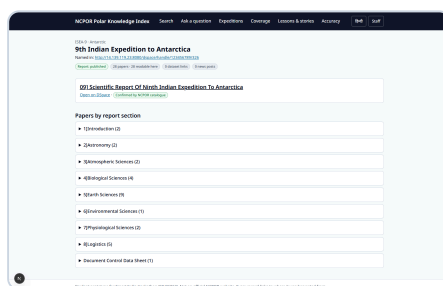
Staff console (login ke saath)

Screen	Kaam
Review queue	Red/amber/green lines, approve/reject/publish
Draft new content	Answer, lesson, announcement banwao
Confirm links	Machine links ko haan/na
Audit log	Sab actions ki history

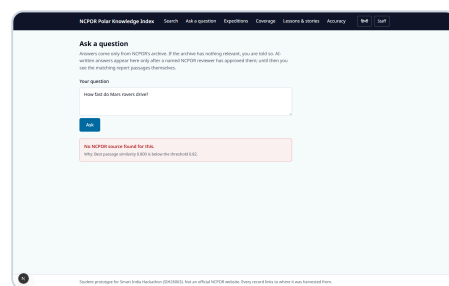
Accessibility: keyboard se chal jaata hai, sahi colour contrast, “Skip to content”, English ↔ हिन्दी toggle — government websites ke **GIGW 3.0 / WCAG 2.1** rules ke hisaab se.



Page viewer: left mein asli scan, right mein padha hua text.



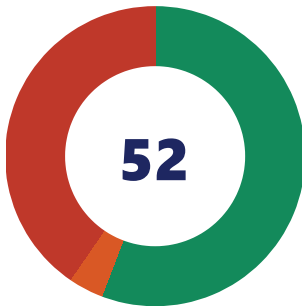
Expedition page (ISEA-9): report aur papers, NCPOR ke sections ke hisaab se.



“How fast do Mars rovers drive?” → **No NCPOR source found**, wajah ke saath.

Coverage view — kya missing hai

Ye feature kisi aur team ke paas nahi. Ye NCPOR leadership ko ek nazar mein batata hai: **kis expedition ka kya public hai aur kya nahi.**



- 29 — public report hai
- 2 — report listed, par "Not Published" (7th, 8th)
- 21 — koi report nahi mili

Kya pata chala?

- DSpace mein reports sirf **1 se 30** tak hain.
- 7th aur 8th** ki report listed hai par "Not Published".
- NCPOR news se **12 baad ke expeditions** (33–37, 40–46) ka pata chala — inki report DSpace mein hai hi nahi.
- Dataset links 0 — kyunki NPDC access NCPOR ki permission ke baad milega.

Imaandari: "—" ka matlab hai "humare harvested sources mein nahi mila", ye nahi ki exist hi nahi karta. Jo expeditions sirf numbering se andaaza hain, unhe saaf "inferred" likha jaata hai.

NCPOR Polar Knowledge Index						
Search	Ask a question	Expeditions	Coverage	Lessons & stories	Accuracy	हिंदी Staff

Expeditions	Public report	Listed, not published	No report found	Dataset link	News or story
52	29	2	21	0	20

Indian Expeditions to Antarctica					
Expedition	Public report	Papers (readable here)	Dataset links	News posts	Approved stories
ISEA-1	published	32 (32)	—	✓ 1	—
ISEA-2	published	24 (0)	—	—	—
ISEA-3	published	25 (0)	—	—	—
ISEA-4	published	18 (18)	—	✓ 1	—
ISEA-5	published	40 (0)	—	✓ 1	—
ISEA-6	published	23 (0)	—	—	—
ISEA-7	listed, not published	—	—	—	—
ISEA-8	listed, not published	—	—	—	—
ISEA-9	published	28 (28)	—	—	—
ISEA-10	published	27 (0)	—	—	—
ISEA-11	published	27 (27)	—	—	—
ISEA-12	published	24 (0)	—	—	—
ISEA-13	published	34 (33)	—	—	—
ISEA-14	published	31 (0)	—	—	—
ISEA-15	published	39 (0)	—	—	—

Asli coverage page: har expedition ek row — public report, papers, dataset links, news, approved stories.

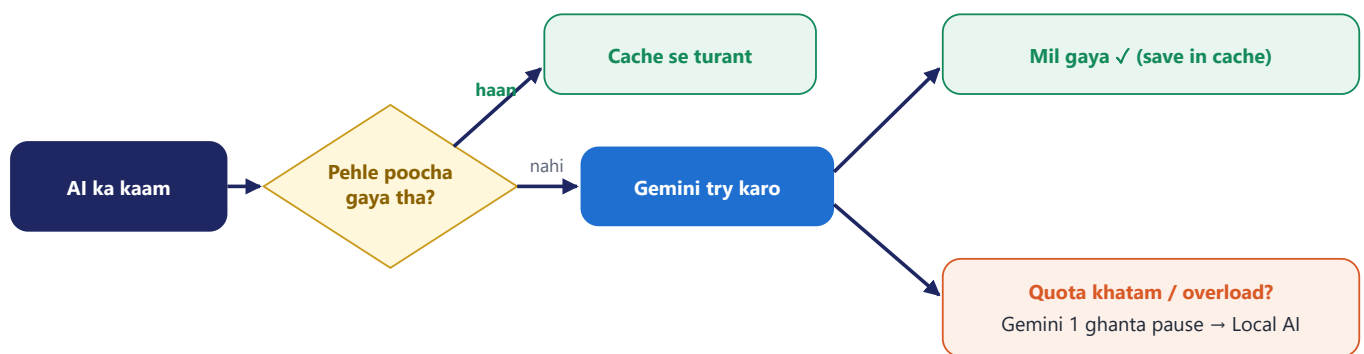
💡 **Simple samjho:** Ye page NCPOR ko batata hai ki **agla outreach effort kahan lagao** — jin expeditions ki koi public kahani nahi, wahan se shuru karo.

Gemini vs Local AI — free setup

Humare system mein AI do jagah se aa sakta hai. Ise **do drivers** ki tarah socho:

	☁ Google Gemini (cloud)	💻 Local model — qwen3:8b via Ollama
Kahan chalta hai	Google ke servers par (internet chahiye)	Humare laptop ke GPU par (RTX 4060)
Quality	Best — sahi citations, achhi Hindi	Theek — citations dheele, checker naram
Limit	Free tier: ~20 requests/din har model	Koi limit nahi
Kharcha	Free tier (billing nahi kiya)	₹0, offline bhi chalta hai
Speed	Kuch second	~2 minute per jawaab (test mein)

Free setup kaise kaam karta hai

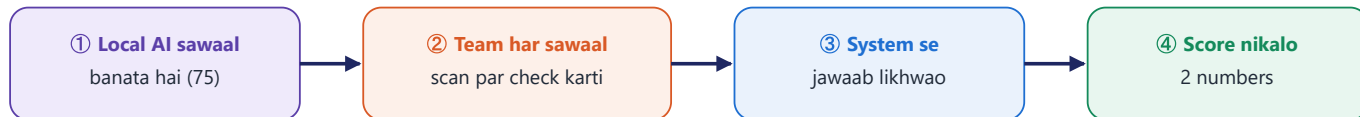


💡 **Simple samjho: Demo trick:** demo wale sawaal pehle se ek baar pooch lo jab Gemini ka quota bacha ho. Unke jawaab cache mein save ho jaate hain — demo ke din internet na bhi ho to bhi wahi achhe jawaab turant dikhenge.

Bulk kaam (papers ko topic dena, test sawaal banana) hamesha local AI karta hai — taaki Gemini ka free quota bacha rahe. Isi tarah humne 44 papers tag kiye aur 75 test sawaal banaye, bilkul free.

Accuracy kaise naapte hain

“Humara AI accurate hai” bolna aasaan hai. Hum **naap ke** batate hain. Iske liye ek **exam paper** banaya jaata hai jiske jawaab insaan ne khud asli pages par dhoondhe hain.



Sirf insaan-verified sawaal score mein count hote hain

Score 1 — Citation accuracy

Jin sawaalon ka jawaab archive mein hai: kitne % jawaabon ne **sahi page** cite kiya jisme **sahi fact** likha hai?

Score 2 — Refusal accuracy

Jin sawaalon ka jawaab archive mein **nahi** hai (Mars, cricket, novels...): kitne % par system ne sahi se **mana kiya**?

Abhi tak ka status

Cheez	Status
Candidate sawaal (local AI se)	75 — 60 answerable (10 reports, 51 papers se), 15 bahar wale
Har candidate ka “expected fact” page par shabd-ba-shabd milna	✓ automatic filter (jo nahi mila, hata diya)
Team verification	Baaki — ~1 ghanta, 2 log
Final scores	[_] — verification ke baad

Candidate sawaalon ke real examples

Sawaal	Expected fact	Faisla
Who led the First Indian Expedition to Antarctica?	“Dr. S. Z. Qasim”	Achha
What was used to collect ice samples from deeper layers?	“a 5 metre long, hand operated stainless steel corer”	Achha
When did the ship sail off from Mauritius?	“1500 hours on 22nd”	Reword — kaunsa expedition?
What are the ages of the samples collected?	“Ages of the samples collected...”	Reject — gol-gol jawaab

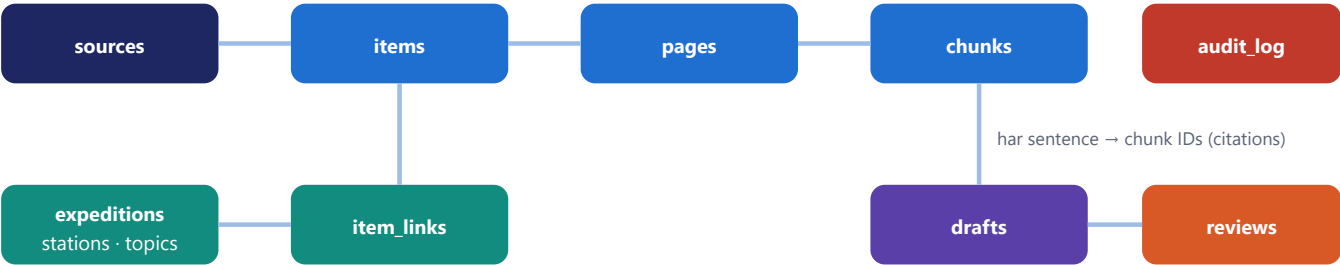
💡 **Simple samjho:** Score jo bhi aaye — 60% ho ya 90% — hum **imaandari se wahi** dikhayenge, failures ke saath. Scientists ki jury ke liye naapa hua imperfect number, bina naape “100% accurate” se zyada bharosemand hai.

Technology — kaunsa tool kya karta hai

Design jaan-boojh ke **chhota aur simple** rakha hai: ek Python backend, ek database, ek website. Student team ise reliably chala sake.

Kaam	Tool	Aasaan bhasha mein
Reports lana	httpx, OAI-PMH, lxml	Internet se records copy karne wale “haath”
PDF padhna	PyMuPDF	PDF kholna, text aur page ki photo nikaalna
Scan padhna	Tesseract OCR 5.4	Photo mein likhe akshar pehchaanna
Data rakhna	PostgreSQL 16 (Docker)	Bada, bharsemand “register” — sab records, links, reviews
Meaning search	pgvector + multilingual-e5	Matlab ko numbers mein badal ke paas wale dhoondhna
Keyword search	PostgreSQL full-text	Exact shabd dhoondhna
AI likhna/check	Google Gemini / Ollama (qwen3:8b)	Cloud ya laptop par chalne wala AI
Backend (dimaag)	Python 3.12, FastAPI	Website aur database ke beech ka “waiter”
Website	Next.js, TypeScript, Tailwind	Jo screen aap dekhte ho
Security	JWT login, bcrypt, roles	Staff login, password safe, kaun kya kar sakta hai
Quality	pytest — 31 tests	Har baar code badle to apne aap jaanch
Code ka ghar	Git + GitHub	Har change ka record, team ke saath share

Data kaise juda hai (simple data model)



18 ROADMAP Abhi kya bacha hai

Kaam	Kaun	Status
Archive: harvest, OCR, linking	Done	✓
Search, cited answers, refusal	Done	✓
AI drafts, fact-checker, Hindi, review	Done	✓
Website (public + staff), production build	Done	✓
Free AI setup (Gemini → local)	Done	✓
GitHub, PPT deck	Done	✓
75 test sawaalon ka verification	Team	Baaki
Evaluation chalana, deck mein scores	Claude (verification ke baad)	Baaki
Demo video + rehearsal	Team	Baaki

Prototype ki seemaayein (honest limits) → future plan

Abhi

- 30 mein se 10 reports poore padhe (baaki ke links hain)
- YouTube videos, photo albums abhi nahi
- NPDC datasets sirf link (permission baaki)
- ~1,000 machine links curator ki haan ka intezaar
- Local AI Gemini se kamzor

Aage

- Saare reports + 31–45 ke digital reports NCPOR se
- Video metadata + photo credits/licence
- NPDC adapter (permission ke baad)
- NCPOR library/NPDC staff curator bane
- Polar & Ocean Museum ke displays ko approved stories bhejna

Demo kaise chalta hai (5–7 minute)

1. DSpace par scanned report dikhao → humare app mein wahi report, expedition se judi, **bina kisi manual entry ke**.
2. Glacier wala sawaal → cited answer → **original page** par click.
3. "Mars rovers" → **mana**.
4. Lesson banao → red line dikhao → reviewer edit/delete karke **apne naam se approve** → English + Hindi publish.
5. Expedition announcement draft → approve.
6. Coverage page → kya missing hai.
7. Accuracy page → dono scores.

Judges ke possible sawaal + jawaab

Q. Ye bhi to ek ChatGPT jaisa chatbot hai?

Nahi. Chatbot apni yaaddasht se kuch bhi bol deta hai. Humara system **sirf NCPOR ke pages** se likhta hai, har line ka page number deta hai, aur jawaab na mile to **mana** karta hai. Aur kuch bhi public hone se pehle insaan approve karta hai.

Q. Agar AI galat bole to?

Teen safety layers: (1) score 0.82 se kam to jawaab hi nahi, (2) fact-checker har line jaanchta hai, red line approval rok deti hai, (3) naam wala reviewer final faisla leta hai. Galti publish ho bhi jaaye to version history ke saath correction hota hai.

Q. NCPOR staff ko kitna extra kaam karna padega?

Data entry: **zero**. System khud harvest karta hai. Staff ka kaam sirf review hai — aur archive linking bina reviewer ke bhi kaam karti hai.

Q. Data kahan se aaya? Real hai?

100% real — NCPOR ka DSpace (OAI-PMH se) aur NCPOR ka news RSS feed. Har record ke saath uska original URL saved hai. Koi dummy data nahi.

Q. Purane scans to kharab honge, OCR galat padhega?

Humne naapa: 10 mein se 9 reports mein 68–92% pages ka text pehle se tha. OCR wale pages ka confidence score dikhta hai, aur har quote ke saath **asli scan** dikhta hai.

Q. Kharcha kitna hai?

Open-source stack, laptop CPU/GPU par chalta hai. AI free tier + local model — licence fee zero.

Q. Hindi kaise?

Approved English draft ka Hindi anuvaad AI banata hai (polar glossary ke saath), uski bhi jaanch aur alag approval hoti hai. Website ka interface English/हिन्दी toggle karta hai.

Q. Doosri teams se alag kya hai?

Asli NCPOR archive, zero data entry, sentence-level fact-check jo approval rokta hai, coverage view, aur **naapi hui accuracy**.

Q. Security?

Public sirf padh sakta hai. Staff login roles ke saath. Passwords encrypted (bcrypt). Audit log ko database level par badla ya mitaaya nahi ja sakta. Secret keys code mein nahi, sirf .env file mein.

Shabd	Aasaan matlab
NCPOR	National Centre for Polar and Ocean Research, Goa — India ke polar expeditions chalane wali sanstha
ISEA	Indian Scientific Expedition to Antarctica — jaise ISEA-9 = 9th expedition
DSpace	NCPOR ki website jahan purani scanned reports rakhi hain
NPDC	National Polar Data Center — scientific datasets ki jagah
Harvesting	Doosri website se records apne aap copy karna
OAI-PMH	Libraries ka standard tareeka catalogue records share karne ka
robots.txt	Website ki file jo batati hai ki bots kya le sakte hain
OCR	Optical Character Recognition — photo mein likhe akshar padhna
Text layer	Scanned PDF ke andar chhupa hua, already padha hua text
Chunk / passage	Text ka chhota tukda (~1,500 akshar), hamesha ek hi page ka
Embedding	Text ka matlab 768 numbers mein — milte matlab = milte numbers
Hybrid search / RRF	Shabd-search + matlab-search ki rankings ko milana
Threshold	Seema-rekha — 0.82 se kam score par jawaab mana
LLM	Large Language Model — Gemini jaisa text likhne wala AI
RAG	Pehle dhoondo, phir sirf usi se likho
Citation	Saboot — kis report ke kis page se baat aayi
Fact-checker / faithfulness	Har line ko uske saboot se milana: supported / partly / not supported
Curator	Staff jo machine links ko haan/na karta hai
Reviewer	Staff jo AI drafts approve/reject/publish karta hai
Audit log	Har action ka record, jo badla ya mitaaya nahi ja sakta
Coverage view	Kis expedition ki kya cheez public hai — ek table mein
Cache	Pehle ke jawaab save — dobara poocho to turant, bina internet
Ollama	Laptop par AI model chalane ka free software
Free tier / quota	Free mein roz ki seemit (limited) requests
API	Programs ke aapas mein baat karne ka darwaaza
GIGW 3.0 / WCAG	Sarkaari websites ke accessibility aur security नियम